

CHAPTER – 1 : SOLUTIONS WORKSHEET

1.	Calculate the molarity of 9.8% (w/w) solution of H_2SO_4 if the density of the solution is 1.02g/mL. Molar mass of $\text{H}_2\text{SO}_4 = 98 \text{ g/mol}$.
2.	What amount of water is required to make 10% ethanol in water solution using 10mL of pure ethanol?
3.	1L of a solution contains 5mL of a certain disinfectant then express this concentration in ppm.
4.	What volume of 95% sulphuric acid (density = 1.85g/mL) and what mass of water must be taken to prepare 100mL of 15% solution of sulphuric acid (density = 1.10 g/mL)
5.	What is the molality of a urea solution in which 0.01g of urea is added to 0.3dm^3 of water at STP?
6.	A solution contains 25% water, 25% ethanol and 50% acetic acid by mass. Calculate the mole fraction of each component.
7.	If the molality of an aqueous solution of cane sugar is 0.4445, what is the mole fraction of cane sugar?
8.	Henry's law constant for carbon dioxide in water is $1.67 \times 10^8 \text{ Pa}$ at 298K. Calculate the quantity of carbon dioxide in 500mL of soda water when packed under 2.5atm CO_2 pressure at 298K.
9.	If the vapour pressure of ethanol at 298K is 40mm of Hg. Its mole fraction in a solution with methanol is 0.8. what will be its vapour pressure in solution, if it obeys Raoult's law?
10.	The vapour pressure of pure liquids A and B at 400K are 450mm and 700mm respectively. Find out the composition of liquid mixture if total vapour pressure at this temperature is 600mm of Hg.
11.	Calculate the vapour pressure of a mixture of 3.9g of benzene and 4.6g of toluene at 298K. Given vapour pressure of pure benzene and toluene are 150mm and 50mm respectively.
12.	A solution is prepared by dissolving 5 g of non-volatile solute in 95 g of water. It has a vapour pressure of 23.375 mm Hg at 25°C . Calculate the molar mass of the solute. (vapour pressure of pure water at 25°C is 23.75 mm Hg).
13.	When 1.5 g of a non-volatile solute was dissolved in 90 g of benzene, the boiling point of benzene raised from 353.23 K to 353.93 K. Calculate the molar mass of the solute. (K_b for benzene = 2.52 kg mol^{-1})

14.	A solution is prepared by dissolving 10 g of non-volatile solute in 200 g of water. It has a vapour pressure of 31.84 mm Hg at 308K. Calculate the molar mass of the solute. (Vapour pressure of pure water at 308 K = 32 mm Hg)
15.	Calculate the mass of compound (molar mass = 256 g mol^{-1}) to be dissolved in 75 g of benzene to lower its freezing point by 0.48 K ($K_f = 5.12 \text{ K kg mol}^{-1}$).
16.	Some ethylene glycol, $\text{HOCH}_2\text{CH}_2\text{OH}$, is added to your car's cooling system along with 5 kg of water. If the freezing point of water glycol solution is -15°C , what is the boiling point of the solution. $K_b = 0.52 \text{ K kg mol}^{-1}$ and $K_f = 1.86 \text{ K kg mol}^{-1}$ for H_2O .
17.	A solution containing 15 g urea (molar mass = 60 g mol^{-1}) per litre of solution in water has the same osmotic pressure (isotonic) as a solution of glucose (molar mass = 180 g mol^{-1}) in water. Calculate the mass of glucose present in one litre of its solution.
18.	18 g of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$ (Molar Mass = 180 g mol^{-1}) is dissolved in 1 kg of water in a sauce pan. At what temperature will this solution boil? (K_b for water = $0.52 \text{ K kg mol}^{-1}$, boiling point of pure water = 373.15 K)
19.	1.00 g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point of benzene by 0.40 K. Find the molar mass of the solute. (K_f for benzene = $5.12 \text{ K kg mol}^{-1}$)
20.	Calculate the amount of KCl which must be added to 1 kg of water so that the freezing point is depressed by 2 K. (K_f for water = $1.86 \text{ K kg mol}^{-1}$)
21.	At 25°C the saturated vapour pressure of water is 3.165 kPa (23.75 mm Hg). Find the saturated vapour pressure of a 5% aqueous solution of urea (carbamide) at the same temperature. (Molar mass of urea = 60.05 g mol^{-1})
22.	A solution of glycerol ($\text{C}_3\text{H}_8\text{O}_3$) in water was prepared by dissolving some glycerol in 500 g of water. This solution has a boiling point of 100.42°C . What mass of glycerol was dissolved to make this solution? (K_b for water = $0.512 \text{ K kg mol}^{-1}$)
23.	What would be the molar mass of a compound if 6.21 g of it dissolved in 24.0 g of chloroform forms a solution that has a boiling point of 68.04°C ? The boiling point of pure chloroform is 61.7°C and the boiling point elevation constant, for chloroform is 3.63°C/m .

24.	Calculate the freezing point of an aqueous solution containing 10.50 g of MgBr_2 in 200 g of water. (Molar mass of $\text{MgBr}_2 = 184 \text{ g mol}^{-1}$) (K_f for water is $1.86 \text{ K kg mol}^{-1}$)
25.	A solution prepared by dissolving 8.95 mg of a gene fragment in 35.0 mL of water has an osmotic pressure of 0.335 torr at 25°C . Assuming the gene fragment is non-electrolyte, determine its molar mass.
26.	At 300 K, 36 g of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$ present per litre in its solution has an osmotic pressure of 4.98 bar. If the osmotic pressure of another glucose solution is 1.52 bar at the same temperature, calculate the concentration of the other solution.
27.	Find the boiling point of a solution containing 0.520 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) dissolved in 80.2 g of water. [Given: K_f for water = 0.52 K/m]
28.	15 g of an unknown molecular substance was dissolved in 450 g of water. The resulting solution freezes at -0.34°C . What is the molar mass of the substance? (K_f for water = $1.86 \text{ K kg mol}^{-1}$)
29.	A solution prepared from 1.25 g of oil of wintergreen (methyl salicylate) in 90.0 g of benzene has a boiling point of 80.31°C . Determine the molar mass of this compound. (Boiling point of pure benzene = 80.10°C and K_b for benzene = $2.53^\circ\text{C kg mol}^{-1}$)
30.	100 mg of a protein is dissolved in just enough water to make 10.0 mL of solution. If this solution has an osmotic pressure of 13.3 mm Hg at 25°C , what is the molar mass of the protein? ($R = 0.0821 \text{ L atm mol}^{-1}\text{K}^{-1}$ and $760 \text{ mm Hg} = 1 \text{ atm}$)
31.	A solution of urea in water has a boiling point of 373.128 K. Calculate the freezing point of the same solution. [Given: For water, $K_f = 1.86 \text{ K m}^{-1}$, $K_b = 0.52 \text{ K m}^{-1}$]